

UNSW MARKETING ANALYTICS HACKATHON 2026

When every story sounds the same

A story does not persuade alone. Its funding speed depends on what surrounds it, when it appears, and why neighbouring stories sound the same.

Team 5 GUYS · 1,453,846 Kiva loans, 2016–2025 · 2025 held out
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2025 funded faster — even as stories became more alike.

MEDIAN FUNDING HOURS • Q1-Q3

2024  121.2 h

2025  68.1 h

–44%

Model sample, Q1-Q3 only, so year-end truncation cannot flatter 2025.

THE SAME YEAR'S MARKET



Volume C

fewer active same-sector loans



Homogeneity H

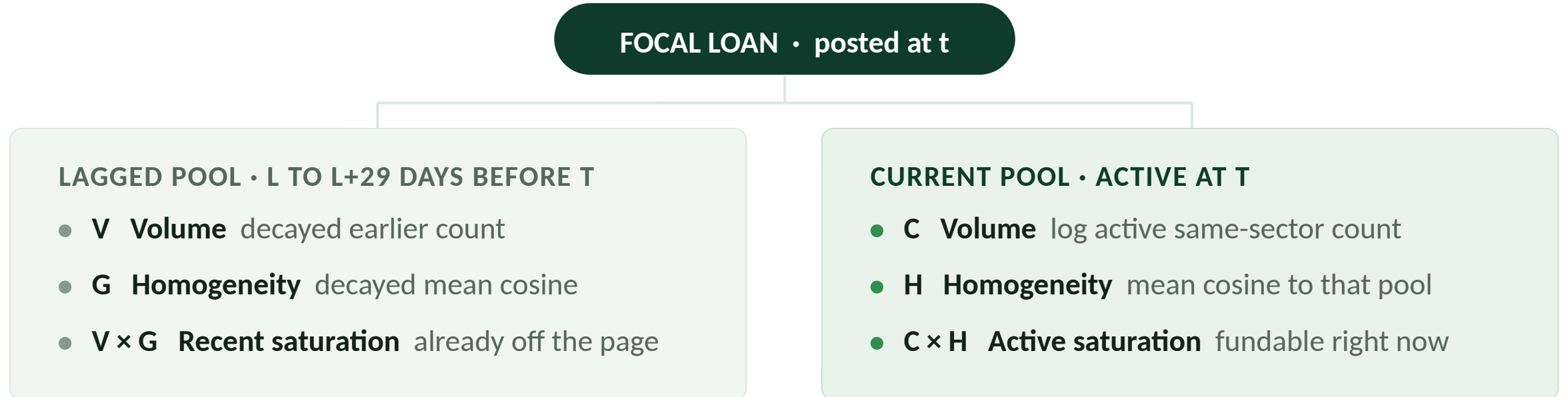
highest mean cosine of the decade

Lower Volume, higher Homogeneity — yet fewer hours.

Similarity alone cannot explain lender response.

KEY METHOD

We rebuilt the choice set each lender could face.



Outcome: FundingHours (fundraisingDate → raisedDate) • 1.23M training loans, 2016–2024 • 2025 held out

Same sector, same week, same country — only the narrative environment differs.

Similarity hurts when it creates live substitutes.

CURRENT POOL • JOINT MOVE IN C AND H

+124% **×2.24**

in conditional FundingHours when Volume C and Homogeneity H move together from P25 to P75

-14.7 pp funded within 72 hours

COST OF +1 SD IN H

+42.5% light market **+60.7%** crowded

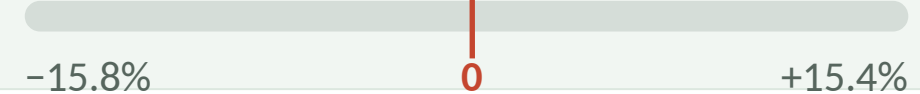
LAGGED POOL • JOINT MOVE IN V AND G

-1.4%

the same P25 → P75 move on V, G and $V \times G$, given the current pool

-0.5 pp funded within 72 hours

95% CONFIDENCE INTERVAL



Live collision is supported. Recent saturation leaves no net drag.

The sameness is partly platform-made — and it bites at the scan stage.

SCAN STAGE • THE SHORT “USE” STATEMENT

+13.6%

more FundingHours moving the use statement from no recurring language to all of it

−3.37 pp funded within 72 hours

On the full description, boilerplate is near-universal — and shows no independent association.

ACTIVE SATURATION $C \times H$

Raw text 0.102 $p = .003$



Residual text 0.052 $p = .129$



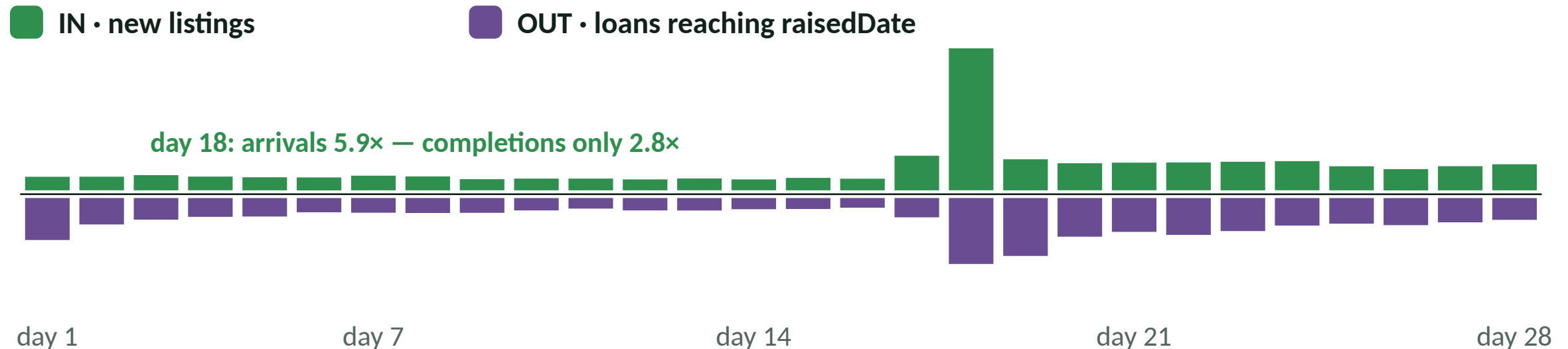
The coefficient roughly halves and loses conventional significance — a shift in magnitude, with no formal cross-model test.

Templates may amplify similarity — but borrowers should not carry the burden.

KEY FINDING • RQ3

Pressure arrives in platform-made pulses: arrivals in, completions out.

Across 2016–2025 the source of similarity moved — similarity at its decade low in 2018, a boilerplate hump in 2020–21, and by 2025 Volume is down while Homogeneity peaks.

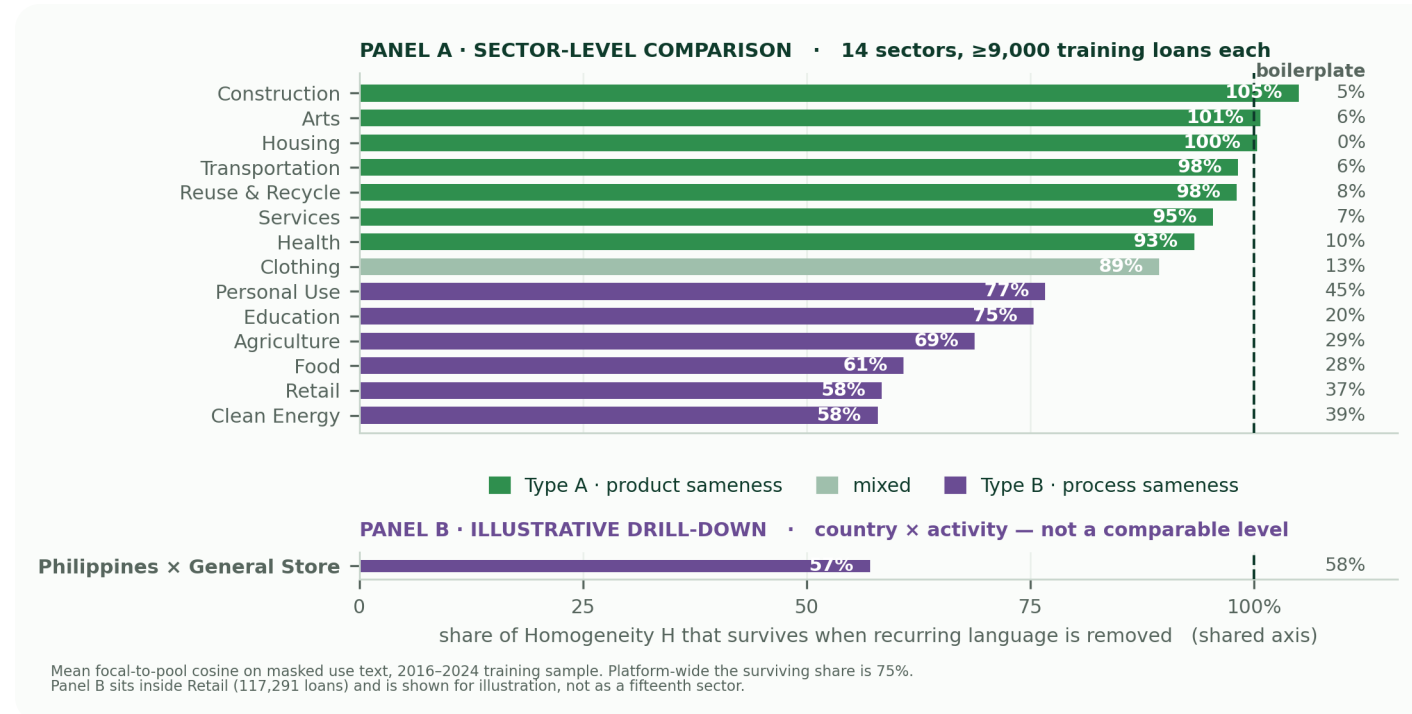


Schematic drawn to the 2024–25 daily index; the fitted series is in Appendix A6.

Both pulses peak on the same day — arrivals spike far higher than completions release.

KEY FINDING • RQ4

Sector-level similarity has different sources.



TYPE A • PRODUCT SAMENESS

Housing, Arts, Transportation — the resemblance survives residual text.

LEVER

Stagger releases • rotate exposure • diversify ranking

TYPE B • PROCESS SAMENESS

Retail, Food, Clean Energy — recurring language carries the resemblance.

LEVER

Partner-approved template variants — not longer borrower stories

Diagnose why stories look alike before choosing the lever.

Panel A is the defended comparison: mean focal-to-pool cosine on raw versus recurring-language-removed text, at sector level.

Panel B is an illustrative drill-down inside Retail — same axis, not a comparable aggregation level.

RECOMMENDATIONS & IMPLICATIONS

Validate first, then operate — the demo is the second half.

PHASE 1 • VALIDATE • BEFORE ANYTHING SHIPS

TEST

Staggered release vs BAU; rotated vs normal exposure; template variants in high-boilerplate markets. Randomised by market-time cell or partner batch.

LEARN

Pre-registered read-out on FundingHours and the 72-hour rate, against fairness, displacement and workload guardrails.

Only levers that clear this gate enter the catalogue.



PHASE 2 • OPERATE • THE LIVE SYSTEM — WHAT THE DEMO SHOWS

SENSE

C, H and the arrivals-minus-completions gap, read by market and by day.

DIAGNOSE

Product sameness, process sameness — or high H with no live substitutes.

ACT

Apply the cleared lever for that diagnosis, or hold. Never at borrower level.

MONITOR

The same metrics and guardrails, running continuously, with a revert rule.

No borrower score. No automatic penalty.

Next: Phase 2, running on the real 2016–2024 data.

Kiva Pulse — Phase 2, the live system.

WHEN

Same market, different day

Housing on day 14: 127 live loans, watch only. Move to day 18 and the pool triples to 298 — the same market becomes an intervention. Pressure is a property of the calendar, not of the borrower.

WHICH KIND

Same H, different problem

Housing and Retail both sit high on Homogeneity. Strip the recurring language and Retail loses 42% of it; Housing loses none. Same reading, opposite lever.

WHAT IT NEVER DOES

It acts on the market, never on the loan

The lever lands on a market-time cell or a Lending Partner's template. No number is attached to a loan, no borrower is asked to rewrite anything, and every action carries a revert rule.

Watch what changes when I move the day — and what never appears on screen.

Four research questions, four answers

Research question	Answer we defend • outcome: FundingHours
RQ1 Active narrative saturation	Yes — $C \times H$ adds funding hours. Volume and Homogeneity reinforce each other.
RQ2 Competition vs wear-out	Signal present, no net drag — the joint restriction rejects, but the components offset.
RQ3 Evolution of the two channels	Not constant — it shifts by year and arrives as listing and completion pulses.
RQ4 Segment boundaries	Not uniform — product sameness and process sameness need different levers.

Pressure depends on when and where a story appears — so the tool must be dynamic.

Data boundaries and the raised-only limit

SAMPLE

- **1,453,846 loans** 1 Jan 2016 – 31 Dec 2025
- **2016–2024** ≈1.23M training loans
- **2025** holdout, never touched
- **Pools** same sector, minimum size ten
- **Text** masked; TF-IDF unigram + bigram

KNOWN LIMITS

- **Raised-only** FundingHours exists only for funded loans
- **No lender identity** wear-out is tested at market level only
- **No partner id** handled by boilerplate decomposition
- **Not borrower quality** never reported at borrower level

These estimates describe funded stories — not whether a story funds at all.

Every row carries a raisedDate, so right censoring cannot be identified and no AFT model is fitted.

The model, and what happens when we change it

$$\begin{aligned} \log(1 + \text{FundingHours}) = & b_0 + b_1 \cdot C + b_2 \cdot H + b_3 \cdot (C \times H) \\ & + b_4 \cdot V + b_5 \cdot G + b_6 \cdot (V \times G) + \gamma \cdot X + FE \end{aligned}$$

Check	Result • C × H and V × G
Precommitted 14-day posting pool	C × H 0.089, p < .001 • V × G 0.026, n.s. — outcome-independent pool agrees
P75-calibrated 16-day posting pool	C × H 0.099, p < .001 • V × G 0.018, n.s.
Pool threshold n ≥ 5 / n ≥ 20	C × H 0.101 / 0.099 • V × G 0.039 / 0.054 — direction and size hold
Residual text	C × H 0.102 → 0.052, loses significance; V × G 0.042 → 0.058, p < .001
Description instead of use	C × H 0.130, p < 1e-6; V × G turns null

Country, activity and week fixed effects; SE clustered by country and week. The five checks are pre-registered specification changes to pool construction, thresholds and text. Half-life frozen at 7 days; specifications are compared side by side, with no formal cross-model test and no forward-window falsification.

RQ2: a rejected joint restriction, and a scenario that nets to zero

THE THREE LAGGED-POOL TERMS

- **V** decayed count, L to L+29 days before t, 7-day half-life
- **G** decayed mean cosine to that same pool
- **V × G** recent saturation — RQ2 tests all three jointly

L = the 95th percentile of funding duration, so lagged loans have nearly all left the page.

WHAT WE REPORT

−1.4%

joint P25 → P75 move in conditional FundingHours

- **95% CI** −15.8% to +15.4%
- **72-hour rate** −0.5 pp
- **Joint scenario** interval covers zero
- **Joint restriction** rejected, p = .002

The lagged channel carries signal — but its components offset, leaving no net drag.

Boilerplate detection, and why the “use” field carries the signal

HOW BOILERPLATE IS IDENTIFIED

- **Rule** 5-grams, ≥500 loans, ≥3 countries
- **Frozen** fixed before any outcome was seen
- **Two uses** covariate, and residual-text variant

MEAN COSINE, RANDOM SAME-SECTOR PAIRS

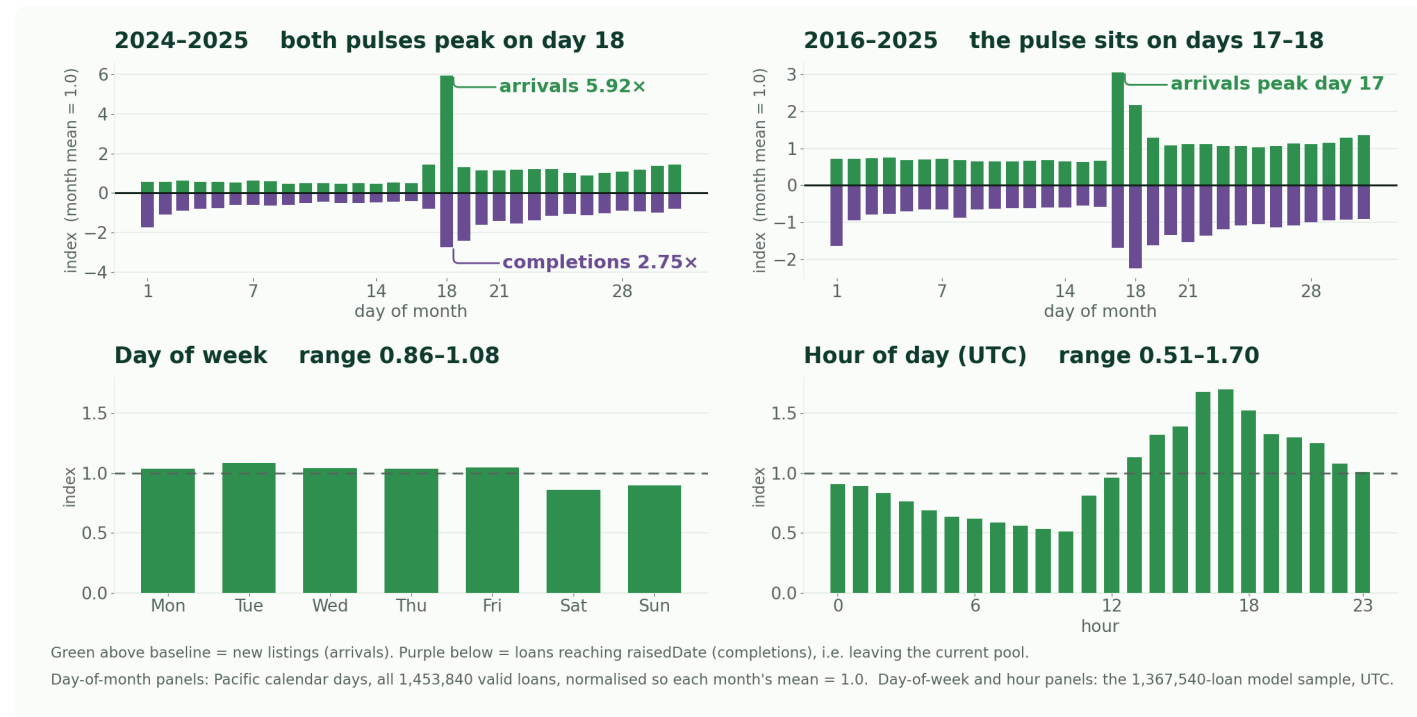


The scan-stage statement separates loans; the description already reads as near-identical.

Specification	C × H	p	Reading
Raw text	0.102	.003	active saturation is present
Residual text	0.052	.129	roughly halves; loses significance
Boilerplate share as covariate	—	n.s.	no separate association given C, H, C × H

Boilerplate appears to work through resemblance to what is on the page at the same time.

The full calendar view: arrivals and completions



WHAT TO POINT AT

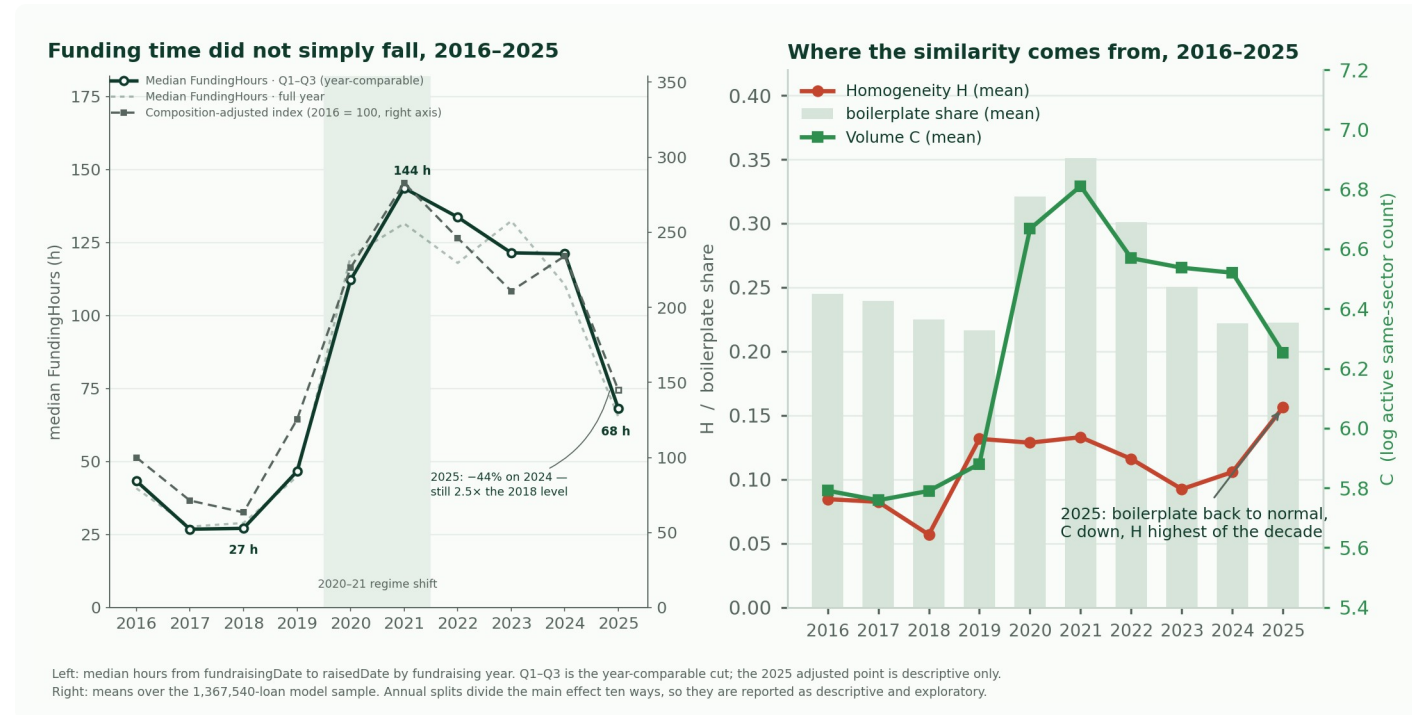
- **Listing pulse** day 18 in 2024-25 (5.9x); day 17 across the decade
- **Completion pulse** peaks on the same day, 2.8x — one day wider
- **Net effect** day 18 carries almost all of the month's net accumulation
- **Week / hour** 0.86-1.08 (Sat low) and 0.51-1.70 — much smaller

Both pulses are supply-side rhythms — not lender mood.

Slide 5 shows a schematic of the mechanism; this is the fitted series.

APPENDIX A7

Year by year: the outcome shifted regime — and the source of similarity moved



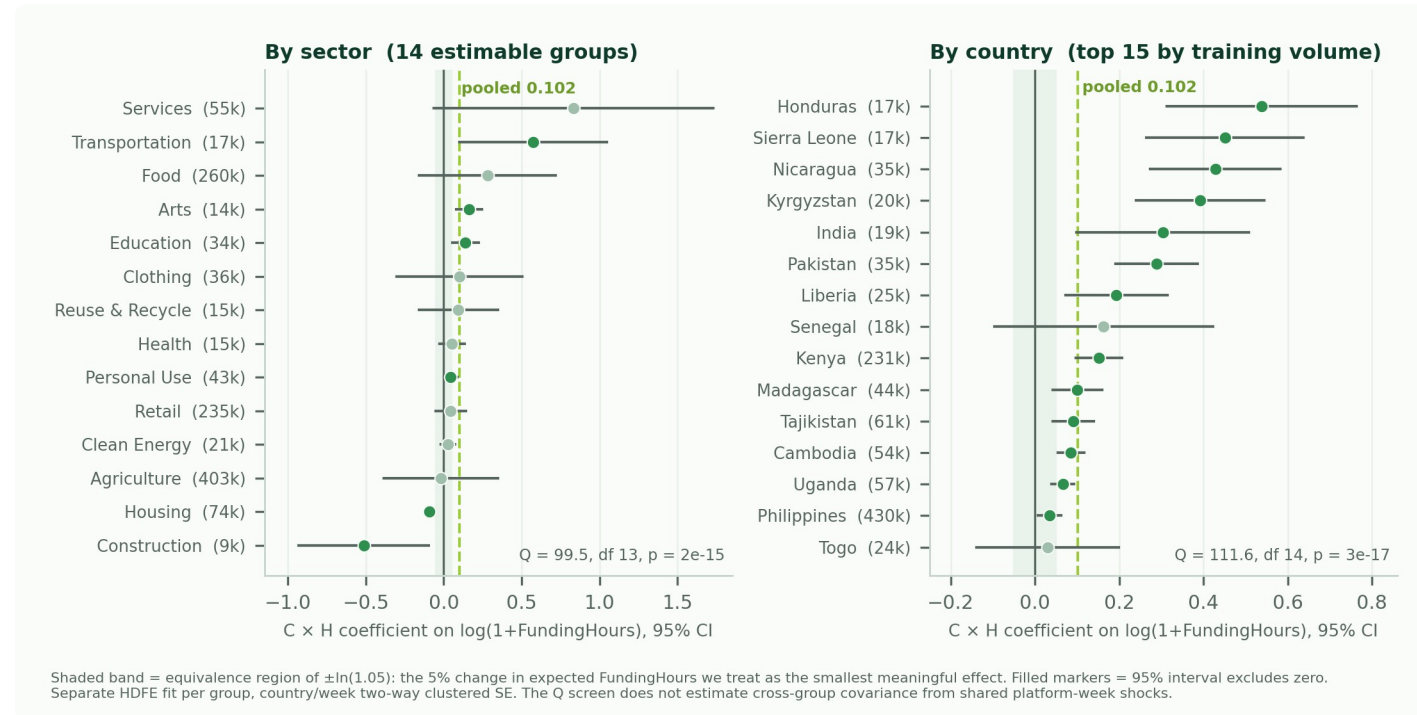
THREE STOPS

- **2016-19** fast market: 27-47 h median; C and H near decade lows
- **2020-21** regime shift: 112 → 144 h; wider pool, boilerplate lifts H
- **2025** partial reversal: 68 h; C down, H highest of the decade

Treat every year label as context, not as a proven cause.

Year-specific $C \times H$ interactions are estimated and reported in the delivered outputs; the annual layer is descriptive and exploratory.

Segments and the boundary of the attention market



THREE CANDIDATE MARKETS

- **Platform-wide** too coarse; absorbed by week FE
- **Active same-sector** our headline definition
- **Country × activity** matches how partners batch

We present the sector definition as a choice, not as a proven boundary.

Joint interaction test first; a 5% change in expected FundingHours is the smallest meaningful effect.

The 2025 holdout — why this cannot be a borrower score

WHAT THE HOLDOUT SHOWED

Adding C, H, $C \times H$, V, G and $V \times G$ did not stably improve prediction of an individual 2025 loan's FundingHours.

- **Market level** signal present and interpretable
- **Loan level** not stable enough to act on

WHAT FOLLOWS FROM IT

- **Do not score borrowers** the holdout does not support a loan-level score
- **Do monitor markets** active saturation is stable as a market state
- **Do randomise** a trigger turns an association into learning

The failure to predict individual loans is why Kiva Pulse is a trigger, not a rating.

Causal boundaries, the data Kiva would need, and sources

CANNOT CLAIM

- **Causal effects** observational pools
- **Individual exposure** no view history
- **Absolute H levels** relative only

LOGS WE WOULD NEED

- **Listing impressions** what was shown, in what order
- **Lender-level views** to test wear-out properly
- **Partner identifier** institution vs narrative

SELECTED SOURCES

Ly & Mason (2012)
Pieters et al. (1999)
Chae et al. (2019)
Hoberg & Phillips (2016) †
Williamson et al. (2021)
Wang (2024) · UNSW (2026)
† read during analysis; not in the submitted proposal

We are not asking Kiva to believe an estimate. We are asking it to run one experiment.

The demo, in two frames: the same market, four days apart

HOUSING • DAY 14 → WATCH ONLY

2

DIAGNOSE

Same H, two different problems — the test is whether H survives removing recurring language

⊖

Are there live substitutes?
Median active same-sector pool ≥ 250 loans on this day

127 in pool

✓

Is the market unusually alike?
Homogeneity percentile ≥ 50

P69 of H

⊖

Where does the sameness come from?
Share of H that survives removing recurring language

not asked

WATCH ONLY

Only 127 comparable loans are live on day 14. Similarity with no live substitutes has no one to lose to — keep it on the dashboard and intervene nowhere.

■ **Why it flips** Nothing about the borrowers changed. The listing batch did — 298 live loans against 127.

HOUSING • DAY 18 → TYPE A • PRODUCT SAMENESS

2

DIAGNOSE

Same H, two different problems — the test is whether H survives removing recurring language

✓

Are there live substitutes?
Median active same-sector pool ≥ 250 loans on this day

298 in pool

✓

Is the market unusually alike?
Homogeneity percentile ≥ 50

P95 of H

✓

Where does the sameness come from?
Share of H that survives removing recurring language

100% survives

TYPE A • PRODUCT SAMENESS

H holds at 100% of its level after recurring language is stripped out (boilerplate share 0%). These are genuinely similar projects shown side by side — the market, not the writing, is the problem.

■ **Retail, day 18** Same high H (P84), but only 58% of it survives residual text — Type B, a template problem, opposite lever.

■ **What it never does** No score is attached to a loan; every read-out ends in a randomised pilot and the metrics to judge it by.

The diagnosis is a property of market-time, not of the borrower.

Phase 2 screenshots. Thresholds are operational choices for triage, not estimated cut-points; no Phase 1 pilot has run, so all levers sit in shadow mode.